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D. Fast Matrix Drift

time limit per test: 1 second

memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

You are given a 2×2 integer matrix A and an integer exponent X ($1 \leq X \leq 10^9$).

Your task is to compute the matrix A^X , where all intermediate operations (additions and multiplications during matrix multiplication) are performed modulo $10^9 + 7$.

Formally, for two matrices:

$$B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}, \quad C = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}.$$

Their product $D = B \times C$ is defined as: $D_{ij} = (b_{i1} \cdot c_{1j} + b_{i2} \cdot c_{2j}) \bmod (10^9 + 7)$.

Input

The first line contains a single integer T ($1 \leq T \leq 10^5$) — the number of test cases.

The first line of each test case contains four integers a_{11} , a_{12} , a_{21} , and a_{22} where $0 \leq a_{ij} \leq 10^9$ — the elements of the matrix A . The second line contains an integer X where $1 \leq X \leq 10^9$.

Output

For each test case, print two lines, each containing two integers: the resulting 2×2 matrix A^X . Each element should be printed modulo $10^9 + 7$.

Examples

input	Copy
2 1 1 1 0 5 2 0 0 2 3	
output	Copy
8 5 5 3 8 0 0 8	
input	Copy
3 1 1 1 1 10 1 2 3 4 12 1 2 1 2 5	
output	Copy

CSE221 Assignment 03 Summer 2026

Contest is running

2 days

Contestant



→ Submit?

Language: Java 21 64bit Choose file: Choose File No file chosen

Success!



Submit

```
512 512
512 512
138067399 201223170
301834755 439902154
81 162
81 162
```

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